

AUGUST/RESIT EXAMINATIONS 2024/2025

MODULE: CSC1018 - Logic

PROGRAMME(S):

CASE	BSc in Computer Applications (Sft.Eng.)
COMSCI	BSc in Computer Science
ECSA	Study Abroad Program EC
ECSAO	Study Abroad Program EC

STAGE: 1,2

EXAMINERS: Dr. David Sinclair (Internal) (Ext:5510)

TIME ALLOWED: 2 hours

INSTRUCTIONS: Answer ALL 6 questions from Section A. Answer 2 out of 3 questions from Section B. Question 1 carries 30 marks. Questions 2 to 9 carry 10 marks.

PLEASE DO NOT TURN OVER THIS PAGE UNTIL INSTRUCTED TO DO SO

The use of programmable or text storing calculators is expressly forbidden.
Please note that where a candidate answers more than the required number of questions, the examiner will mark all questions attempted and then select the highest scoring ones.

There are no additional requirements for this paper.

Section A

QUESTION 1

[Total marks: 30]

If Bob has spare time he is happy, If Bob is busy, he doesn't have time to be sad.

1(a) [10 Marks]

Convert the statements above into a set of *propositional* formulae.

Hint: Bob can be happy or sad, but not both at the same time. Bob can be busy or have spare time, but not both at the same time.

1(b) [10 Marks]

Convert the predicate formulae from Question 1(a) into *Conjunctive Normal Form (CNF)*. Clearly describe each step in the conversion.

1(c) [10 Marks]

Using Resolution, prove that Bob is not sad.

[End Question 1]

QUESTION 2

[Total marks: 10]

[10 Marks]

Without using truth tables, show that

$$(A \rightarrow C) \wedge (B \rightarrow C) \equiv (A \vee B) \rightarrow C$$

Clearly indicate which logical equivalence is used at each step.

[End Question 2]

QUESTION 3

[Total marks: 10]

[10 Marks]

Using semantic tableaux, prove the following propositional formula is valid.

$$(p \rightarrow q) \rightarrow ((p \rightarrow \neg q) \rightarrow \neg p)$$

Clearly indicate which inference rule is used at each step.

[End Question 3]

QUESTION 4**[Total marks: 10]**

[10 Marks]

Using the Gentzen System $\mathcal{G}$ prove

$$\vdash (A \vee (B \wedge C)) \rightarrow ((A \vee B) \wedge (A \vee C))$$

Clearly indicate which inference rule is used at each step.

[End Question 4]**QUESTION 5****[Total marks: 10]**

[10 Marks]

Using *semantic tableaux* prove the following first-order statement is valid.

$$\vdash (\forall x.p(x) \vee \forall x.q(x)) \rightarrow \forall x.(p(x) \vee q(x))$$

Clearly indicate which inference rule is used at each step.

[End Question 5]**QUESTION 6****[Total marks: 10]**

[10 Marks]

Using Gentzen system $\mathcal{G}$, prove the following predicate formula.

$$\forall x.(p(x) \rightarrow q(x)) \rightarrow (\forall x.p(x) \rightarrow \forall x.q(x))$$

Clearly indicate which inference rule is used at each step.

[End Question 6]

Section B

QUESTION 7

[Total marks: 10]

[10 Marks]

Consider the following grid of points.

a	-	b	-	c	-	d
e	-	f	-	g	-	h
i	-	j	-	k	-	l
m	-	n	-	o	-	p

This grid can be represented by the following relations:

- `nexteast(X,Y)` is true if X is the immediate east point of Y, e.g. `nexteast(g,f)`.
- `nextsouth(X,Y)` is true if X is the immediate south point of Y, e.g. `nextsouth(m,i)`.

Given these relations, write the following Prolog relations:

- `south(X,Y)` is true if X can be reached from Y by only travelling directly south.
- `east(X,Y)` is true if X can be reached from Y by only travelling directly east.
- `southoreast(X,Y)` is true if X can be reached from Y by only travelling directly south or east.

[End Question 7]

QUESTION 8

[Total marks: 10]

[10 Marks]

Write Prolog relations to:

- sum the elements of a list of integers; and
- find the maximum of a list of elements.

[End Question 8]

QUESTION 9

[Total marks: 10]

[10 Marks]

Describe *cuts* in Prolog. Why are *cuts* used? Illustrate your answer with examples.

[End Question 9]

APPENDICES

Semantic Tableaux

α	α_1	α_2	β	β_1	β_2
$\neg\neg A_1$	A_1		$\neg(B_1 \wedge B_2)$	$\neg B_1$	$\neg B_2$
$A_1 \wedge A_2$	A_1	A_2	$B_1 \vee B_2$	B_1	B_2
$\neg(A_1 \vee A_2)$	$\neg A_1$	$\neg A_2$	$B_1 \rightarrow B_2$	$\neg B_1$	B_2
$\neg(A_1 \rightarrow A_2)$	A_1	$\neg A_2$	$B_1 \uparrow B_2$	$\neg B_1$	$\neg B_2$
$\neg(A_1 \uparrow A_2)$	A_1	A_2	$\neg(B_1 \downarrow B_2)$	B_1	B_2
$A_1 \downarrow A_2$	$\neg A_1$	$\neg A_2$	$\neg(B_1 \leftrightarrow B_2)$	$\neg(B_1 \rightarrow B_2)$	$\neg(B_2 \rightarrow B_1)$
$A_1 \leftrightarrow A_2$	$A_1 \rightarrow A_2$	$A_2 \rightarrow A_1$	$B_1 \oplus B_2$	$\neg(B_1 \rightarrow B_2)$	$\neg(B_2 \rightarrow B_1)$
$\neg(A_1 \oplus A_2)$	$A_1 \rightarrow A_2$	$A_2 \rightarrow A_1$			

γ	$\gamma(a)$	δ	$\delta(a)$
$\forall x.A(x)$	$A(a)$	$\exists x.A(x)$	$A(a)$
$\neg\exists x.A(x)$	$\neg A(a)$	$\neg\forall x.A(x)$	$\neg A(a)$

Gentzen Logic

$$\frac{\vdash U'_1 \cup \{\alpha_1, \alpha_2\}}{\vdash U'_1 \cup \{\alpha\}} \quad \frac{\vdash U'_1 \cup \{\beta_1\} \quad \vdash U'_2 \cup \{\beta_2\}}{\vdash U'_1 \cup U'_2 \cup \{\beta\}}$$

$$\frac{\vdash U'_1 \cup \{\gamma, \gamma(a)\}}{\vdash U'_1 \cup \{\gamma\}} \quad \frac{\vdash U'_1 \cup \{\delta(a)\}}{\vdash U'_1 \cup \{\delta\}}$$

$\frac{A}{B}$ is a *sequent* and means that if A is true, then B can be inferred.

α	α_1	α_2	β	β_1	β_2
$\neg\neg A$	A		$B_1 \wedge B_2$	B_1	B_2
$\neg(A_1 \wedge A_2)$	$\neg A_1$	$\neg A_2$	$\neg(B_1 \vee B_2)$	$\neg B_1$	$\neg B_2$
$A_1 \vee A_2$	A_1	A_2	$\neg(B_1 \rightarrow B_2)$	B_1	$\neg B_2$
$A_1 \rightarrow A_2$	$\neg A_1$	A_2	$\neg(B_1 \uparrow B_2)$	B_1	B_2
$A_1 \uparrow A_2$	$\neg A_1$	$\neg A_2$	$B_1 \downarrow B_2$	$\neg B_1$	$\neg B_2$
$\neg(A_1 \downarrow A_2)$	A_1	A_2	$B_1 \leftrightarrow B_2$	$B_1 \rightarrow B_2$	$B_2 \rightarrow B_1$
$\neg(A_1 \leftrightarrow A_2)$	$\neg(A_1 \rightarrow A_2)$	$\neg(A_2 \rightarrow A_1)$	$\neg(B_1 \oplus B_2)$	$B_1 \rightarrow B_2$	$B_2 \rightarrow B_1$
$A_1 \oplus A_2$	$\neg(A_1 \rightarrow A_2)$	$\neg(A_2 \rightarrow A_1)$			

γ	$\gamma(a)$	δ	$\delta(a)$
$\exists x.A(x)$	$A(a)$	$\forall x.A(x)$	$A(a)$
$\neg\forall x.A(x)$	$\neg A(a)$	$\neg\exists x.A(x)$	$\neg A(a)$

[END OF APPENDICES]

[END OF EXAM]